Luigui **Gallardo-Becerra**

Bioinformatician | Molecular Biologist

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Experienced Bioinformatician and Molecular Biologist seeking new challenges to apply and expand my expertise in computational biology, data science, and molecular research. Over the past several years, I have contributed to diverse research projects utilizing modern genomic technologies (NGS), advanced data analysis, and data science methodologies to generate novel biological insights that have resulted in high-impact peer-reviewed scientific publications.

Research Experience

Graduate Research Assistant (Data Science & Bioinformatics) — *Institute of Biotechnology, UNAM*

January 2019 - July 2025

I contributed to multiple research projects, from initial data acquisition and generation to final reporting and publication. The key responsibilities I executed include:

- **Research project management:** Participated in defining primary and secondary objectives and research questions for various projects to generate novel biological insights valuable to the scientific community
- **High-performance computing (HPC) administration:** Managed laboratory servers, installed Linux operating systems and specialized software/packages, optimized server performance, and implemented secure computing environments
- **Bioinformatics pipeline development:** Created automated pipelines to streamline repetitive workflows and ensure reproducible analyses using Bash, Python, Snakemake, and Rabix Composer
- **Software development:** Developed multiple software tools to address specific research challenges using Bash, Python, Perl, and R
- **Data visualization:** Created comprehensive plots and visualizations using R, RStudio, Jupyter Notebooks, Excel, GraphPad, and Tableau
- **Version control and CI/CD:** Maintained GitHub repositories and implemented continuous integration workflows
- **Scientific writing and communication:** Authored weekly and biannual reports for laboratory members using Markdown and Word. Co-authored peer-reviewed publications that achieved high citation rates in their respective fields
- **Scientific outreach:** Delivered presentations to general audiences and academic conferences, participated in scientific dissemination events, specialized seminars, and international congresses

Education

National Autonomous University of Mexico (UNAM) — *Mexico City, Mexico*

Ph.D. in Computational Biology — *January 2019 - July 2025*

National Autonomous University of Mexico (UNAM) — *Mexico City, Mexico*

Master of Science in Computational Biology — *August 2016 - January 2019*

University of Guadalajara (UDG) — *Guadalajara, Mexico*

Bachelor of Science in Molecular Biology — *August 2012 - January 2016*

Teaching Experience

- Guest lecturer at Universidad de La Sabana (Chia, Colombia). December 2023.
- Microbiota Symposium from Theory to Clinical Practice, Colegio de Profesionales de la Nutrición de Querétaro y el Bajío. June 2021.
- Bioinformatics and Reproducible Research Workshop, Postgraduate Program in Biological Sciences, UNAM. October 2019.
- Guest lecturer at Center of Genomic Sciences, UNAM. August 2018.

Directed Theses

- Bachelor's Thesis: Itzel Abigail Hernández Reyna, Bachelor's in Nutrition (Universidad Autónoma del Estado de Morelos), December 2019. "Impact of bacteriophages associated with childhood obesity on the intestinal metagenome."

Soft Skills

- I am a fast learner with strong adaptability to new technologies and methodologies. When faced with unfamiliar concepts, I proactively seek learning opportunities to master them.
- I possess excellent problem-solving capabilities and can effectively manage significant workloads across multiple projects from diverse domains without compromising quality or deadlines.
- I am an effective team player with strong communication skills, capable of working independently.
- I have expertise in server maintenance, GitHub repository management, and script development for research projects.

Programming Languages, Tools & Bioinformatics Skills:

- **Programming Languages:** Python, C#, R, SQL, Bash, HTML & CSS, JavaScript, TypeScript
- **Web Frameworks:** Django, ASP.NET, Angular, Shiny
- **Data Science & Engineering Tools:** High-Performance Computing (HPC), Linux, Tableau, Docker, Snakemake, Jupyter Notebooks, RStudio, MySQL, PostgreSQL, MongoDB
- **Bioinformatics Skills:** Data curation and database creation, 16S rRNA sequencing analysis and profiling, Genome and transcriptome assembly and annotation, Metagenomic analysis, Metatranscriptomic analysis, Viral metagenomics (viromics)

Languages

- English – Full professional proficiency
- Spanish – Native

Publications

1. **Gallardo-Becerra L**, Cornejo-Granados F, Romero-Hidalgo S, Lopez-Zavala AA, Cota-Huízar A, Cervantes-Echeverría M, et al. Host genome drives the microbiota enrichment of beneficial microbes in shrimp: exploring the hologenome perspective. Animal Microbiome. 2025 May 22;7(1):50.
2. Manzo R, **Gallardo-Becerra L**, Díaz de León-Guerrero S, Villaseñor T, Cornejo-Granados F, Salazar-León J, et al. Environmental Enrichment Prevents Gut Dysbiosis Progression and Enhances Glucose Metabolism in High-Fat Diet-Induced Obese Mice. International Journal of Molecular Sciences. 2024 Jan;25(13):6904.
3. Jatuyosporn T, Laohawutthichai P, Romo JPO, **Gallardo-Becerra L**, Lopez FS, Tassanakajon A, et al. White spot syndrome virus impact on the expression of immune genes and gut microbiome of black tiger shrimp Penaeus monodon. Sci Rep. 2023 Jan 18;13(1):996.
4. **Gallardo-Becerra L**, Cervantes-Echeverría M, Cornejo-Granados F, Vazquez-Morado LE, Ochoa-Leyva A. Perspectives in Searching Antimicrobial Peptides (AMPs) Produced by the Microbiota. Microb Ecol. 2023 Dec 1;87(1):8.
5. Chino de la Cruz CM, Cornejo-Granados F, **Gallardo-Becerra L**, Rodríguez-Alegría ME, Ochoa-Leyva A, López Munguía A. Complete genome sequence and characterization of a novel Enterococcus faecium with probiotic potential isolated from the gut of Litopenaeus vannamei. Microbial Genomics. 2023;9(3):000938.
6. Cervantes-Echeverría M, **Gallardo-Becerra L**, Cornejo-Granados F, Ochoa-Leyva A. The Two-Faced Role of crAssphage Subfamilies in Obesity and Metabolic Syndrome: Between Good and Evil. Genes. 2023 Jan;14(1):139.
7. Palomino-Hermosillo YA, Berumen-Varela G, Ochoa-Jiménez VA, Balois-Morales R, Jiménez-Zurita JO, Bautista-Rosales PU, et al. Transcriptome Analysis of Soursop (Annona muricata L.) Fruit under Postharvest Storage Identifies Genes Families Involved in Ripening. Plants (Basel). 2022 Jul 7;11(14):1798.
8. Ochoa-Romo JP, Cornejo-Granados F, Lopez-Zavala AA, Viana MT, Sánchez F, **Gallardo-Becerra L**, et al. Agavin induces beneficial microbes in the shrimp microbiota under farming conditions. Sci Rep. 2022 Apr 16;12(1):6392.
9. Bikel S, **Gallardo-Becerra L**, Cornejo-Granados F, Ochoa-Leyva A. Protocol for the isolation, sequencing, and analysis of the gut phageome from human fecal samples. STAR Protocols. 2022 Mar 18;3(1):101170.

10. Bikel S, López-Leal G, Cornejo-Granados F, **Gallardo-Becerra L**, García-López R, Sánchez F, et al. Gut dsDNA virome shows diversity and richness alterations associated with childhood obesity and metabolic syndrome. iScience. 2021 Aug 20;24(8):102900.
11. **Gallardo-Becerra L**, Cornejo-Granados F, García-López R, Valdez-Lara A, Bikel S, Canizales-Quinteros S, et al. Metatranscriptomic analysis to define the Secrebiome, and 16S rRNA profiling of the gut microbiome in obesity and metabolic syndrome of Mexican children. Microb Cell Fact. 2020 Dec;19(1):61.
12. **Gallardo-Becerra L**, Cornejo-Granados F, Leonardo-Reza M, Ochoa-Romo JP, Ochoa-Leyva A. A meta-analysis reveals the environmental and host factors shaping the structure and function of the shrimp microbiota. PeerJ. 2018 Aug 10;6:e5382.
13. Cornejo-Granados F, Lopez-Zavala AA, **Gallardo-Becerra L**, Mendoza-Vargas A, Sánchez F, Vichido R, et al. Microbiome of Pacific Whiteleg shrimp reveals differential bacterial community composition between Wild, Aquacultured and AHPND/EMS outbreak conditions. Sci Rep. 2017 Dec;7(1):11783.